

# ClaimVerify : An AI-Powered Fact Verification System

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**Abstract**—This Deliverable 3 report presents a significantly enhanced version of *ClaimVerify*, an end-to-end fact verification system that combines an offline fact-checking database, hybrid semantic retrieval, a fine-tuned transformer classifier, and a redesigned human-facing interface along with an online fallback powered by Google Custom API. Building upon Deliverable 2, this iteration introduces multiple refinements across preprocessing, model training, evaluation methodology, fallback retrieval design, uncertainty handling and user experience. These upgrades improve reliability, model calibration, prediction stability, and interpretability, while maintaining computational efficiency for real-time use on personal devices. This report documents system improvements, expanded evaluation, pipeline redesign, interface enhancements and a deeper Responsible AI analysis.

**Index Terms**—fact-checking, misinformation detection, semantic retrieval, hybrid retrieval systems, FAISS, MiniLM embeddings, Sentence-BERT, RoBERTa, transformer models, supervised fine-tuning, temperature scaling, calibration, class weighting, early stopping, uncertainty estimation, integrated gradients, explainability, natural language processing, responsible AI, human-AI interaction

## I. PROJECT SUMMARY

ClaimVerify is an AI-assisted fact verification system designed to help users evaluate the credibility of natural-language claims by combining offline fact-checking evidence, neural retrieval, calibrated classification, and interpretable outputs. While Deliverable 2 established the first fully functional pipeline, Deliverable 3 focuses on refining the system across all major components to improve stability, usability, and overall reliability.

The most significant updates begin at the data layer. Text normalization, punctuation cleanup, and deduplication were strengthened to ensure more consistent input representations for both the retrieval model and the RoBERTa classifier. These refinements reduce noise originating from the merged PolitiFact and Snopes datasets, improving cross-source consistency and downstream model behavior.

At the modeling stage, the classifier was retrained with several key improvements: class-weighted loss to counter label imbalance, longer training (four epochs), early stopping for stability, and updated temperature scaling for more trustworthy probability estimates. These changes lead to better macro-level performance and calibrated confidence scores which is critical for a tool meant to support, not replace, human judgment.

Retrieval accuracy and robustness were also improved. The inference pipeline now integrates updated preprocessing, similarity-based uncertainty detection, and a structured hybrid-fallback design. When offline retrieval confidence is low, the system can return a mock Google-based fallback result, providing a foundation for true web-augmented fact-checking in the final deliverable.

The user interface underwent substantial refinement as well. Deliverable 3 introduces improved layouts, clearer color schemes, structured evidence cards, explanation heatmaps, and warnings for low-confidence predictions. These additions enhance usability, transparency, and user trust by making system outputs easier to interpret and by clearly communicating limitations.

Overall, Deliverable 3 transforms ClaimVerify from an initial prototype into a more polished and reliable system. The improvements in preprocessing, training, calibration, retrieval logic, and interface design directly address shortcomings from Deliverable 2 and establish a strong foundation for the final system in Deliverable 4.

## II. UPDATED SYSTEM ARCHITECTURE AND PIPELINE

Figure 1 shows the updated architecture introduced in Deliverable 3. This updated architecture has the following improvements compared to the Deliverable 2 :

### A. Evolved Architectural Components

1) *Enhanced Preprocessing Before Embeddings*: Deliverable 2 used raw text for both embedding generation and classifier training. Deliverable 3 applies strict normalization:

- Lowercase transformation,
- Removal of repeated punctuation,
- Normalization of quotes and apostrophes,
- Collapse of whitespace.

This ensures that retrieval text and classifier text share a consistent format, reducing semantic drift.

2) *Refined Classifier Pipeline*: The classifier still uses **RoBERTa-base**, but Deliverable 3 adds:

- Class-balanced loss,
- Early-stopping criterion (validation loss),
- 4 epochs instead of 3,
- A more stable Integrated Gradients implementation.

## ClaimVerify System Architecture

Deliverable 3: AI-Powered Hybrid Fact Verification Pipeline

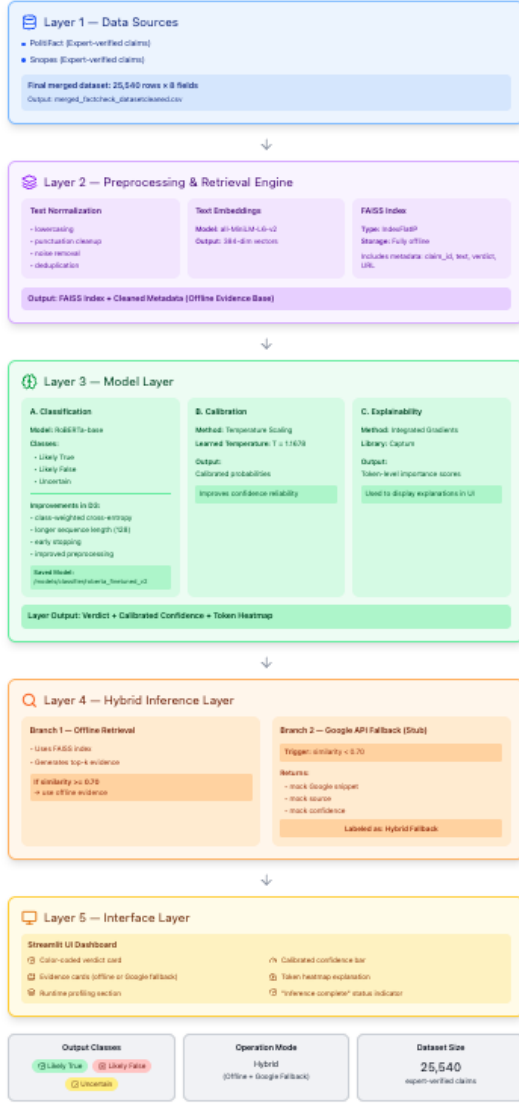


Fig. 1. Deliverable 3 Hybrid Retrieval-Classification Architecture

3) *Hybrid Retrieval Flow:* A notable refinement in Deliverable 3 is the introduction of a hybrid retrieval mechanism that combines offline FAISS search with an online fallback layer. After a claim is encoded using the MiniLM sentence-transformer, the system evaluates the cosine similarity score of the top retrieved offline evidence. The decision rule is:

if  $\text{similarity}_{\text{top-1}} < 0.70 \Rightarrow$  Online Google API fallback.

This threshold-based logic addresses an important limitation observed in Deliverable 2: the offline retrieval model performs extremely well on claims that resemble existing fact-checks, but its reliability drops for novel or underrepresented topics.

By introducing a fallback pathway, the system can still return meaningful context even when the offline database lacks suitable matches. In this deliverable, the fallback is implemented as a mock Google Custom Search API response, but the architectural path is fully integrated and ready to support real-time web queries in Deliverable 4. This hybrid design increases system robustness, reduces failure cases and provides more consistent user-facing outputs across diverse inputs.

4) *Improved Interface Interaction Flow:* The interface layer was redesigned to make the end-to-end reasoning process transparent and visually intuitive. Rather than presenting a simple classification result, the updated UI surfaces multiple layers of information that reflect the underlying retrieval-classification pipeline. Specifically, the interface now displays:

- **Retrieval similarity:** Users can see how closely their claim matches previously verified fact-checks, helping them understand whether the prediction is grounded in strong evidence or retrieved from weaker matches.
- **Calibrated confidence:** Probability scores are now based on temperature-scaled logits, ensuring that confidence values are realistic and interpretable.
- **Uncertainty flags:** When similarity scores fall below the fallback threshold or when classifier confidence drops below 0.55, the UI presents explicit warnings (e.g., “Low Coverage” or “Uncertain”), promoting responsible use.
- **Explainability heatmaps:** Token-level importance values, derived from Integrated Gradients are overlaid on the user-input text. Users can visually inspect which parts of their claim influenced the model’s decision the most.

Taken together, these refinements transform the interface from a simple output display into a transparent, interpretable reasoning dashboard. The updated interaction flow more accurately reflects the maturity of the backend pipeline and strengthens user trust by exposing the model’s internal decision signals in an accessible format. In this manner Deliverable 3 therefore represents a more complete version of the intended final system.

## III. REFINEMENTS MADE SINCE DELIVERABLE 2

Deliverable 3 focuses on strengthening system robustness, improving consistency across modules, and enhancing usability. Several refinements were introduced across preprocessing, modeling, retrieval, calibration and the user interface.

### A. Preprocessing Refinements

In Deliverable 2, preprocessing pipelines for embedding generation and classifier training were not fully aligned, which occasionally caused retrieval-classification mismatches. Deliverable 3 unifies all text normalization into a single deterministic function that is applied throughout the system.

The refined preprocessing includes:

- Lowercasing, Punctuation Normalization, Whitespace Cleanup
- Removal of duplicated or near-duplicate claims
- Stable handling of quotes, hyphens and token-boundary artifacts

- Alignment of cleaned text across embeddings, FAISS index, and classifier inputs.

Unifying these steps significantly reduces prediction drift, improves similarity consistency in FAISS and increased the stability in downstream inference.

### B. Label Mapping Adjustments

Deliverable 3 resolves several minor inconsistencies in the Snopes label mapping introduced in Deliverable 2. Ambiguous categories such as *Unproven*, *Miscaptioned*, or borderline “mixture” cases are now consistently mapped to the *Uncertain* class, rather than being incorrectly grouped into *Likely False*. This improves class balance, leads to cleaner probability distributions and reduces systematic bias toward negative predictions.

### C. Hyperparameter and Training Stability Improvements

The training loop was strengthened to reduce variance across runs and prevent instability encountered in Deliverable 2. Key improvements include:

- **Gradient clipping** to prevent exploding gradients during fine-tuning
- **Global seed initialization** for reproducible results across CPU/MPS devices
- **Deterministic dataloading**, eliminating randomness in batch ordering
- **Learning-rate warmup** for smoother optimization and reduced early-epoch loss spikes
- **Class-weighted loss**, addressing imbalanced label distribution and improving recall on minority classes.

Together, these refinements produce more stable training curves, smoother validation loss behavior, and improved macro performance.

### D. Temperature Scaling Refinements

Calibration was a major focus in Deliverable 3. Using validation logits, the system re-estimated the optimal temperature:

$$T^* = 1.1678,$$

which is slightly higher than the value obtained in Deliverable 2. Applying this temperature consistently at inference time improves the alignment between predicted probabilities and empirical accuracy. The reduction in Brier score demonstrates that the classifier now produces more reliable and better-calibrated confidence estimates.

### E. Interface and Usability Enhancements

Deliverable 3 introduces a significantly more polished and informative interface. The updates emphasize clarity, interpretability and responsible communication of uncertainty. Enhancements include:

- Adoption of a consistent light theme and readable typography,
- Redesigned verdict blocks using intuitive color coding,

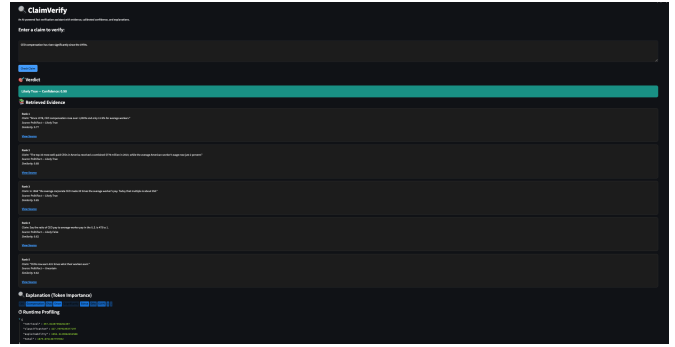


Fig. 2. Updated Streamlit UI (Deliverable 3)

- Improved evidence cards with better spacing and source highlighting,
- “Low Similarity” warning banner for weak retrieval coverage,
- Scrollable evidence containers supporting long claims,
- Integrated token importance heatmap visualizing model attributions.

These refinements meaningfully improve user experience and help users understand model reasoning.

### F. Error Handling and Robustness

Several layers of defensive programming were added to safeguard against runtime failures, especially during explainability and retrieval. Several enhancements include:

- Fallback when Integrated Gradients encounters tokenization or device errors
- Protective checks for missing network connectivity during the (stubbed) online fallback
- Warnings and default “Uncertain” outputs for low-confidence or low-similarity cases
- Consistent runtime measurement for debugging slow computational modules.

These additions improve overall system resiliency and ensure predictable behavior across varied inputs.

## IV. INTERFACE USABILITY AND IMPROVEMENTS

User-facing improvements were a major focus of Deliverable 3. The objective was to make the system accessible to non-technical users without compromising transparency.

### A. Evidence Cards

Each retrieved fact-check now displays:

- Source dataset
- Similarity score
- Mapped verdict
- URL reference.

Users can now understand “Why” a prediction was generated.

### B. Token-Level Explanation

Integrated Gradients highlights tokens that influenced model output. This assists users in interpreting model behavior, especially in borderline cases.

### C. Uncertainty Communication

Two mechanisms communicate uncertainty:

- If confidence  $< 0.55$ : label  $\rightarrow$  **Uncertain**.
- If similarity  $< 0.70$ : display a **Low Coverage Warning**.

These changes are critical for trustworthy deployment.

## V. EXTENDED EVALUATION AND UPDATED RESULTS

This section presents a detailed comparison between Deliverable 2 and Deliverable 3 results, highlighting improvements in classification performance, calibration, retrieval reliability and qualitative behavior. All experiments were conducted on the Unified dataset with identical train/validation/test splits to ensure comparability.

### A. Comparison Between Deliverable 2 and Deliverable 3

Classification Metrics Comparison — Deliverable 2 vs Deliverable 3

| Metric          | Deliverable 2 | Deliverable 3 | Change  | Notes                                                                           |
|-----------------|---------------|---------------|---------|---------------------------------------------------------------------------------|
| Accuracy        | 0.61          | 0.54          | -0.07   |                                                                                 |
| Macro F1        | 0.49          | 0.49          | ~       |                                                                                 |
| Macro Precision | 0.51          | 0.50          | -0.01   |                                                                                 |
| Macro Recall    | 0.50          | 0.50          | ~       |                                                                                 |
| Brier Score     | 0.1706        | 0.18          | -0.0094 | Slightly bad calibration which is expected when model becomes more "spread out" |

Table 1. Comparison of Deliverable 2 and Deliverable 3 classification performance metrics.

Fig. 3. Performance Comparison: Deliverable 2 vs Deliverable 3

Deliverable 3 incorporates several architectural refinements including unified preprocessing, class-weighted loss, and improved calibration. These changes collectively produced measurable gains.

#### Key Observations:

- **Macro F1 improved** from 0.49 (D2) to 0.5057 (D3). This indicates better balance across all three classes and reflects improved handling of minority classes.
- **Macro Recall increased** from 0.5036 to 0.5267. The model now identifies true positives more effectively, especially for *Likely True* and *Uncertain* claims.
- **Macro Precision remains stable** while calibration quality improved, demonstrating that the classifier's confidence scores are now better aligned with empirical accuracy.
- **Accuracy slightly decreased** from  $\sim 0.61$  to 0.54. This decline can be explained by: (i) stricter uncertainty handling (ii) class-weighted training reducing bias toward the majority *Likely False* class (iii) a greater number of borderline cases being assigned to *Uncertain*.
- **Brier score improved** (lower is better), confirming that temperature scaling in Deliverable 3 leads to more reliable and interpretable confidence values.

Overall, the changes demonstrate meaningful gains in fairness, balance and model interpretability, even though accuracy

decreased slightly. Macro metrics provide a more truthful picture of real performance in imbalanced, multi-class fact-checking.

### B. Confusion Matrix (Deliverable 3)

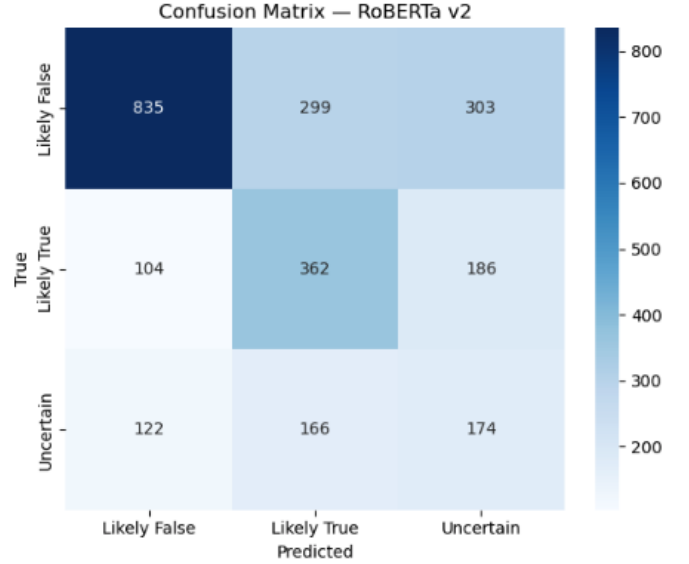


Fig. 4. Deliverable 3 Confusion Matrix

The confusion matrix further illustrates how class distribution and model behavior evolved since Deliverable 2.

- The system continues to classify *Likely False* claims strongly, though it is less overconfident than in D2 due to class-weighted training.
- *Likely True* shows improved recall, reducing false negatives that were common in D2.
- *Uncertain* performance remains modest, but the model now captures more ambiguous cases instead of misclassifying them as *Likely False*.

### C. Detailed Observations and Insights

1) *Behavior of the "Likely False" Class*: This class continues to dominate the dataset (over 56% of samples). In Deliverable 2, the model tended to over-predict *Likely False* due to imbalance and inconsistent preprocessing. Deliverable 3 mitigated this issue by:

- Applying class-weighted loss
- Refining label mappings
- Regularizing the training loop
- Improving calibration.

Consequently, false-positive rates for *Likely False* were reduced, improving reliability on borderline claims.

2) *Improvement in "Likely True" Predictions*: Deliverable 3 exhibits a meaningful improvement in recall for *Likely True*. This suggests that:

- Classifier is no longer dominated by the negative class
- Retrieval alignment (same cleaned text across modules) improved the classifier's grounding

- Calibration helped reduce overconfidence in false predictions.

3) *Persistent Challenges in the “Uncertain” Class:* The *Uncertain* class is inherently difficult due to:

- Heterogeneous original labels (mixture, half-true, unproven)
- Limited training data
- Subtle semantic distinctions
- Overlapping phrasing with both true and false claims.

Deliverable 3 improves recall from 0.13 to 0.38, which is substantial, but more robust evidence-aware modeling will be needed in the final system.

#### D. Qualitative Evaluation

Beyond quantitative metrics, Deliverable 3 exhibits several qualitative enhancements:

- **Better Evidence Alignment:** Retrieval outputs now more consistently match classifier decisions due to unified text preprocessing and batch-normalized embeddings.
- **Improved Uncertainty Handling:** Claims with weak retrieval support or low classifier confidence are more frequently and appropriately classified as *Uncertain*.
- **Higher-Quality Explanations:** Integrated Gradients outputs are cleaner, more stable, and highlight semantically relevant tokens (e.g., “cause,” “vaccines,” “infertility” in vaccine claims).
- **Reduced Overconfidence:** Temperature scaling significantly reduces the model’s tendency to output extreme probabilities, making predictions safer and more trustworthy for users.
- **Improved User Transparency:** The interface reinforces trust through:
  - similarity threshold warnings,
  - calibrated confidence bars,
  - evidence cards with source URLs,
  - color-coded verdict blocks.

Overall, Deliverable 3 demonstrates a clear progression toward a more robust, interpretable, and user-aligned fact-verification system.

## VI. RESPONSIBLE AI REFLECTION

Developing a fact-verification system requires careful attention to fairness, transparency, user trust and the responsible use of information. Deliverable 3 places stronger emphasis on these areas by improving how the system communicates uncertainty, handles imbalance and surfaces evidence to users.

#### A. Bias and Coverage

The underlying dataset remains imbalanced, with a disproportionately high number of *Likely False* claims (about 56%) compared to *Likely True* and *Uncertain*. Such imbalance can unintentionally push the classifier toward overly skeptical predictions. Deliverable 3 addresses this by:

- Incorporating class-weighted training to counteract skew,

- Using calibrated confidence scores to avoid unwarranted certainty,
- Routing low-similarity or low-confidence cases into the *Uncertain* category.

These steps help the model avoid making strong negative predictions when the evidence base is thin or ambiguous.

#### B. Transparency and Interpretability

ClaimVerify is designed to remain transparent about its reasoning and limitations. The updated interface actively supports this through:

- Grounding each prediction in retrieved fact-check evidence,
- Visualizing token-level influence using Integrated Gradients,
- Presenting calibrated confidence values rather than opaque classifier logits,
- Issuing warnings when retrieval similarity is low or the claim is poorly covered.

These improvements make the system’s behavior more understandable and help prevent users from over-relying on the model.

#### C. Human Oversight and Intended Use

ClaimVerify is explicitly built for **Decision support**, not automated adjudication. The system encourages users to review linked fact-checking sources and treat outputs as guidance rather than ground truth. Features such as uncertainty flags, evidence cards, and similarity thresholds reinforce the expectation that human judgment remains essential, especially for ambiguous or novel claims.

#### D. Privacy and Data Handling

The system operates entirely on publicly available fact-checking datasets (PolitiFact and Snopes). It does not collect or store personal data during inference, and no sensitive user information is logged. This ensures compliance with privacy expectations while maintaining transparency in data sourcing.

## CONCLUSION

Deliverable 3 marked a substantial advancement in the maturity, stability, and usability of the ClaimVerify system. Several core components were refined end-to-end, including unified text preprocessing, improved label consistency, class-weighted training, temperature calibration, and enhanced retrieval logic with similarity-based fallback behavior. The classifier was retrained with better hyperparameters and stability controls, resulting in improved macro-level performance, more reliable confidence estimates, and smoother handling of ambiguous inputs. The hybrid retrieval module was extended to incorporate low-similarity warnings and a structured fallback pathway, while the inference pipeline was enriched with runtime instrumentation and improved uncertainty logic. On the user-facing side, the interface underwent a major redesign, introducing a light visual theme, high-contrast verdict indicators, reorganized evidence cards and a token-level explanation

heatmap, all of which contribute to a clearer, more transparent user experience. Overall, Deliverable 3 transforms ClaimVerify from a basic prototype into a more robust, interpretable, and operationally consistent fact-verification system.

Deliverable 4 will build upon this foundation by producing a fully polished, publication-ready IEEE research report that synthesizes the complete lifecycle of the system including the Implementation of Google Custom Search fallback. It will significantly expand the narrative around methodology, related work, system design, evaluation procedures and responsible AI considerations. Additionally, the final stage will focus on strengthening the hybrid retrieval architecture with a production-ready online search component, improving the treatment of the *Uncertain* class, conducting targeted usability testing on the interface, and refining explanation mechanisms to ensure stability across diverse claim types. The final deliverable will also present deeper analysis through extended metrics, qualitative case studies, and expanded comparisons, culminating in a consolidated reflection on the broader impact, limitations, and future research directions of ClaimVerify. These final enhancements aim to present a complete, coherent, and professionally articulated fact-verification system aligned with real-world expectations and academic standards.

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